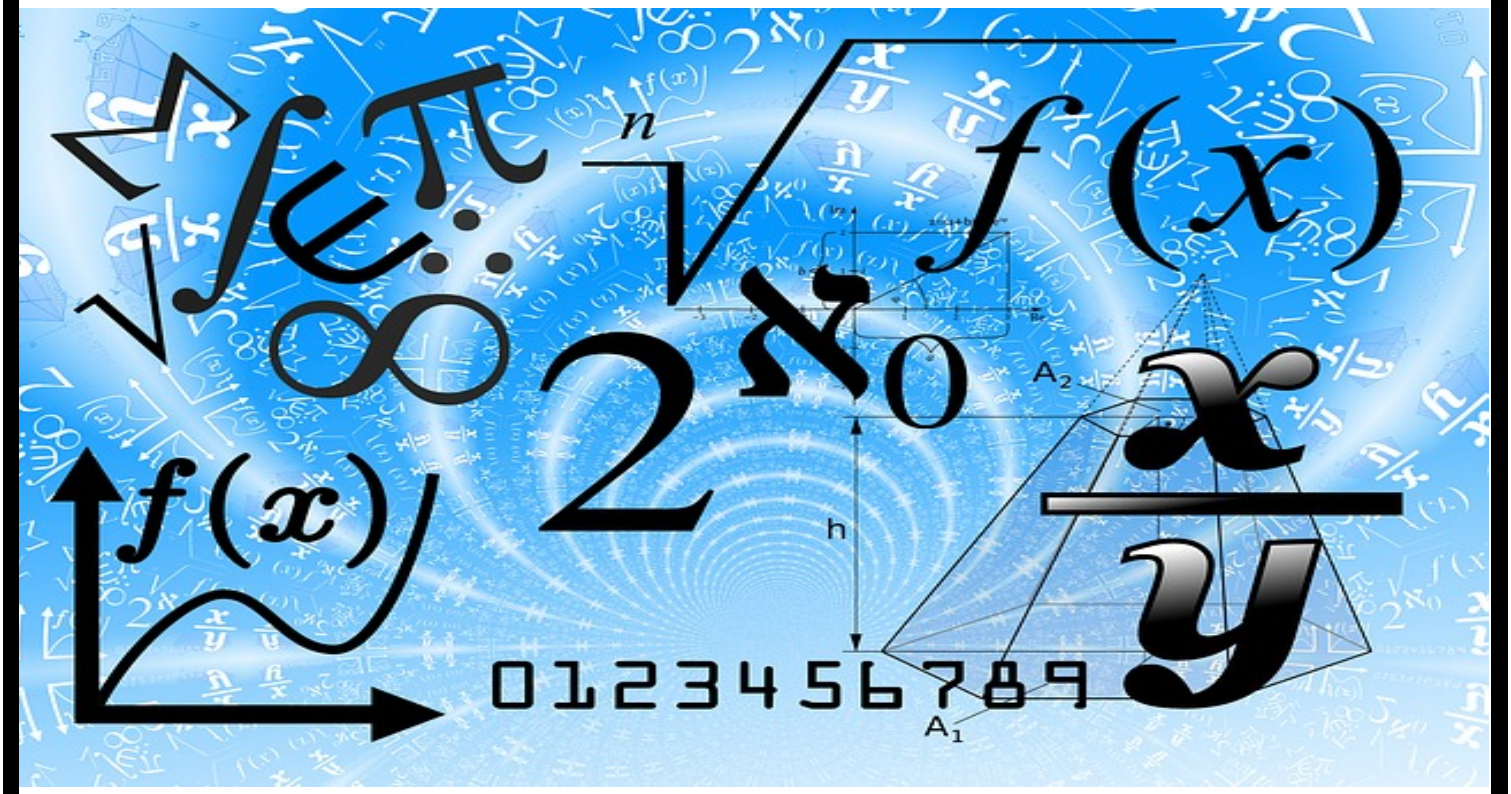
THE VIRTUAL MEET EXPERIENCE

2025-2026

HS VIRTUAL CHALLENGE MEET #4



MATHEMATICS

DO NOT OPEN TEST UNTIL TOLD TO DO SO

The Virtual Challenge Meets™

1. Coraline's mother just gave her \$95 to buy dinner for her family. She told Coraline to keep any change leftover. Caroline went to the local Greek restaurant. She found a deal that allowed her to buy 4 gyros for \$35.50 altogether. She also bought 3 salads for \$12.95 each and a tray of baklava for \$10.50. If the tax rate was 8.25% and the restaurant would not accept a tip, how much did Coraline get to keep for herself?

(A) \$3.00 (B) \$3.05 (C) \$3.10 (D) \$3.15 (E) \$3.20

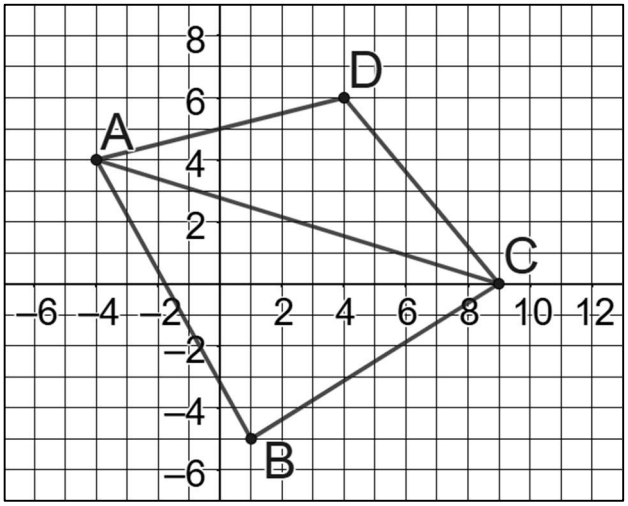
2. For a fundraiser, the AHS math team hired a professional catering service for a pancake breakfast. The members of the math team sold student tickets for \$6 and adult tickets for \$10. The team sold a total of 144 tickets and raised a total of \$1,080. If the catering service cost \$200, how many adult tickets did they sell?

(A) 92 (B) 96 (C) 100 (D) 104 (E) 108

3. Gerald is taking Algebra 2 with Mrs. Courtney. His semester grade is weighted as follows: 15% homework, 20% quizzes, and 65% tests. His homework average is 95, and his quiz average is 83. There are four tests this semester; his first three test scores are 82, 96, and 84. What is the minimum score he must earn on the fourth test to have at least an 89.5 semester average?

(A) 95 (B) 96 (C) 97 (D) 98 (E) 99

4 – 8. Consider polygon ABCD as shown to the right. All marked points have integer coordinates.



4. The point $(a,8)$ lies on the perpendicular bisector of side $\overline{CD}$. What is the value of a? (nearest tenth)

(A) 12.5 (B) 12.7 (C) 12.9
(D) 13.1 (E) 13.3

5. What is the area of triangle ADC?

(A) 28.5 (B) 29.0 (C) 29.5
(D) 30.0 (E) 30.5

6. What is the length of the longest median of triangle ADC? (nearest tenth)

(A) 10.3 (B) 10.4 (C) 10.5 (D) 10.6 (E) 10.7

7. What is the measure of angle ABC? (nearest degree)

(A) 84° (B) 85° (C) 86° (D) 87° (E) 88°

8. Segment AD is one side of equilateral triangle ADE. Point E (x,y) is above point D such that $y > 0$. What is the value of $x+y$? (nearest tenth)

(A) 9.8 (B) 9.9 (C) 10.0 (D) 10.1 (E) 10.2

9. Desmond baked a large batch of brownies to share with his family and friends. He took one-third of the brownies to give to everyone in his first period class. He then gave half of the brownies to his friend Nico. Then, he gave three-quarters of the remaining brownies to his parents. If he had 9 brownies left to himself, how many brownies did he share with his first period class?

(A) 36 (B) 38 (C) 40 (D) 42 (E) 44

10. Consider the quadratic given by $f(x) = -2x^2 + bx + c$ with zeros at $x = -3$ and $x = -\frac{1}{2}$. What is the value of $f(1)$?

(A) -14 (B) -13 (C) -12 (D) -11 (E) -10

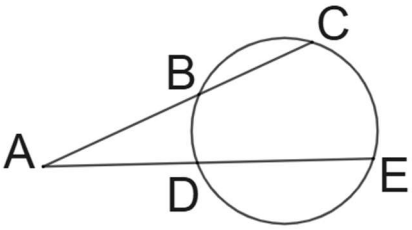
11. Usain Bolt holds the world record for the fastest 200-meter race with a time of 19.19 seconds. What was Usain Bolt's average speed to set this record? (nearest tenth mile per hour)

(A) 22.7 mph (B) 22.9 mph (C) 23.1 mph (D) 23.3 mph (E) 23.5 mph

12. Consider the function $g(x)$ such that $g(x) + 3 \cdot g(4 - x) = 3 - x$. Find the value of $g(4)$.

(A) $\frac{1}{4}$ (B) $\frac{1}{2}$ (C) $\frac{3}{4}$ (D) 1 (E) $\frac{5}{4}$

13 – 14. Consider the circle with secants $\overline{AC}$ and $\overline{AE}$. The length of $\overline{AE}$ is 41.6, the length of $\overline{AC}$ is 38.5 and the length of $\overline{AB}$ is 1.52 units longer than $\overline{AD}$.



13. What is the length of $\overline{AD}$? (nearest tenth)

(A) 18.7 (B) 18.9 (C) 19.1 (D) 19.3 (E) 19.5

14. If the radius of the circle is approximately 11.9, what is the area of minor segment BC? (nearest tenth)

(A) 52.3 (B) 52.5 (C) 52.7 (D) 52.9 (E) 53.1

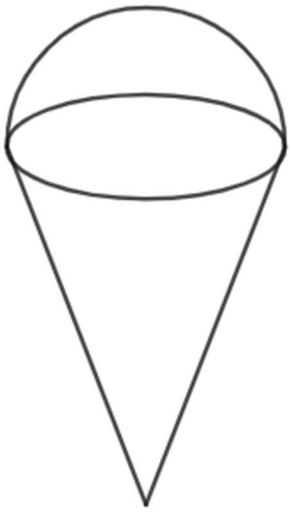
15. If $s(x)$ is the oblique asymptote of $h(x) = \frac{x^3 - 9}{x + 2}$, then the value of $s(3)$ is _____.

(A) 7 (B) 8 (C) 9 (D) 10 (E) 11

16. Rick has samples of two different types of radioactive material. Sample A has a half-life of 28 hours, and sample B has a half-life of 92 hours. Rick receives the two samples at the same time. Sample A is initially 56 grams and sample B is initially 18 grams. How long after Rick received the samples will their mass be the same? (nearest tenth hour)

(A) 65.7 hours (B) 65.9 hours (C) 66.1 hours (D) 66.3 hours (E) 66.5 hours

17 – 19. Consider the huge ice cream cone as shown to the right in the shape of an inverted cone with a hemisphere of ice cream on top. Assume the cone is completely packed with ice cream underneath the hemisphere. The straight-line distance from the vertex of the cone to the top of the hemisphere is 8 inches and the height of the cone is twice the radius of the hemisphere.



17. If the thickness of the cone is negligible, what is the total volume of the ice cream? (nearest tenth pint)

(A) 2.6 pints (B) 2.7 pints (C) 2.8 pints (D) 2.9 pints (E) 3.0 pints

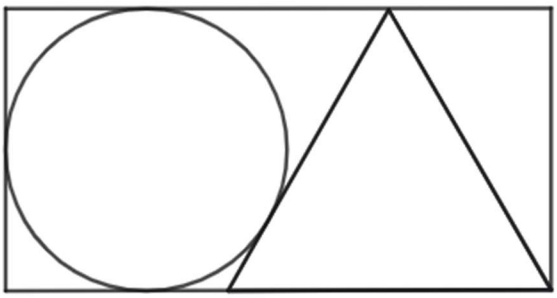
18. What is the vertex angle of the cone? (nearest tenth degree)

(A) 52.5° (B) 52.7° (C) 52.9° (D) 53.1° (E) 53.3°

19. The shop that makes this enormous ice cream cone has 12 different flavors for customers to choose from. This particular ice cream cone takes three massive scoops to fill. How many unique ice cream cones can customers make with the three scoops?

(A) 220 (B) 256 (C) 292 (D) 328 (E) 364

20 – 21. The image to the right shows a circle that is tangent to the rectangle and to the equilateral triangle to the right. The height of the rectangle is 11 units.



20. What is the side length of the equilateral triangle? (nearest tenth)

(A) 12.5 units (B) 12.6 units (C) 12.7 units
(D) 12.8 units (E) 12.9 units

21. What is the area that is inside the rectangle but outside of the circle and triangle? (nearest tenth)

(A) 70.2 units^2 (B) 70.3 units^2 (C) 70.4 units^2 (D) 70.5 units^2 (E) 70.6 units^2

22. The number that is $3/8$ of the way from A to $5\frac{2}{5}$ is $\frac{17}{30}$. What is the value of A?

(A) -3 (B) $-2\frac{5}{6}$ (C) $-2\frac{2}{3}$ (D) $-2\frac{1}{2}$ (E) $-2\frac{1}{3}$

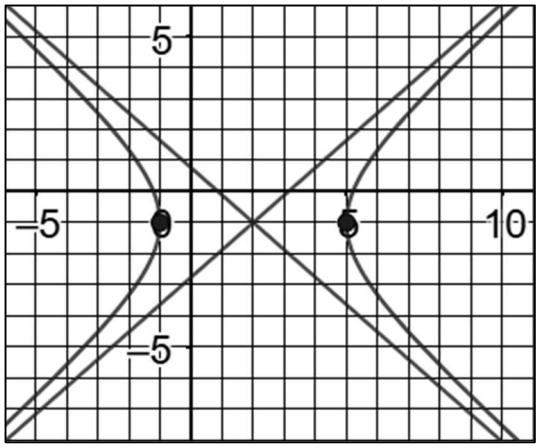
23. Alan and Vincent are painting the interior of a house. Alan can paint 400 square feet in 3 hours and Vincent can paint 100 square feet in 30 minutes. The house they were hired to paint has an area of 3,258 square feet that needs to be painted. Vincent shows up at 8 AM and begins work. Alan shows up at 9:15 AM and begins work. If they continue working without taking a break, what time will they finish painting the house? (nearest minute)

(A) 6:16 PM (B) 6:18 PM (C) 6:20 PM (D) 6:22 PM (E) 6:24 PM

24. Consider the sphere given by $x^2+y^2+z^2-4x+6y-12z-18=0$. What is the area of the largest cross section of the sphere? (nearest integer)

(A) 195 (B) 200 (C) 205 (D) 210 (E) 215

25 – 26. Consider the graph of the hyperbola as shown to the right. The marked points have integer coordinates, and the slope of the asymptotes is $\pm\frac{\sqrt{7}}{3}$.



25. The equation of the hyperbola is $\frac{(x-h)^2}{a^2}-\frac{(y-k)^2}{b^2}=1$. What is the value of $a+b+h+k$? (nearest hundredth)

(A) 6.61 (B) 6.63 (C) 6.65
(D) 6.67 (E) 6.69

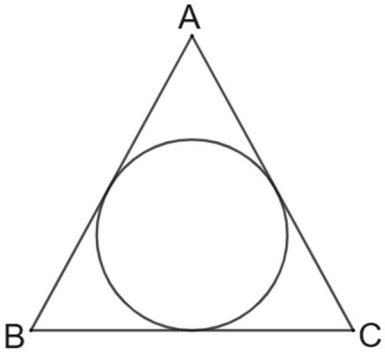
26. The point $(x,10)$ lies on the graph of the hyperbola such that $x>0$. What is the value of x? (nearest hundredth)

(A) 14.79 (B) 14.80 (C) 14.81 (D) 14.82 (E) 14.83

27. Consider the functions $f(x)=3^x-2$ and $g(x)=2^x-1$. If $h(x)=\frac{f(x)}{g(x)}$, then $h^{-1}(-2)=$ ____. (nearest thousandth)

(A) 0.341 (B) 0.342 (C) 0.343 (D) 0.344 (E) 0.345

28 – 29. Consider isosceles triangle ABC as shown to the right with an inscribed circle. The measure of angle A is $58°$ and the area of the circle is 210.



28. What is the length of side AB? (nearest hundredth)

(A) 28.63 (B) 28.65 (C) 28.67 (D) 28.69 (E) 28.71

29. What percentage of the total area of the figure is inside the triangle but outside of the circle? (nearest tenth percent)

(A) 39.6% (B) 39.8% (C) 40.0% (D) 40.2% (E) 40.4%

30. Consider the vectors $u=\langle -2,a,6\rangle$ and $v=\langle a-1,3,1\rangle$. The angle between the vectors is $78.214°$. What is the value of a if $a>0$? (nearest integer)

(A) 5 (B) 6 (C) 7 (D) 8 (E) 9

31 – 32. Consider the graph of the conic given by $2x^2 - 5xy + 6y^2 - 30 = 0$.

31. The angle of rotation of the conic is _____. (nearest tenth)

(A) $25.5°$ (B) $25.7°$ (C) $25.9°$ (D) $26.1°$ (E) $26.3°$

32. This conic can be classified as a/an _____.

(A) 2 parallel lines (B) parabola (C) hyperbola (D) ellipse (E) circle

33. Which of the following is a y-intercept of the graph of the polar function $r = 4 - 2.3\sin\theta$ expressed in rectangular form?

(A) $(0,-6.3)$ (B) $(0,-6.2)$ (C) $(0,-6.1)$ (D) $(0,-6.0)$ (E) $(0,-5.9)$

34. The Burj Khalifa is currently the world's tallest skyscraper with a height of 828 meters. The Lounge is located at a height of 585 meters within the Burj Khalifa. How far onto the horizon can a person see when they looking out from The Lounge? (nearest tenth mile)

(A) 53.3 miles (B) 53.7 miles (C) 54.1 miles (D) 54.5 miles (E) 54.9 miles

35. Consider the sequence given by $2,5,11,19,30,44,m,n,\ldots$. What is the value of $m+n$?

(A) 143 (B) 145 (C) 147 (D) 149 (E) 151

36. The temperature of an exoplanet varies sinusoidally with the time of day. The length of the exoplanet's day is 62 hours. At $t = 0$, the planet has a low temperature of $-42°C$ and at $t = 31$, the planet has a high temperature of $210°C$. How many hours in a day does the planet have a temperature greater than $60°C$? (nearest tenth)

(A) 34.6 hours (B) 34.7 hours (C) 34.8 hours (D) 34.9 hours (E) 35.0 hours

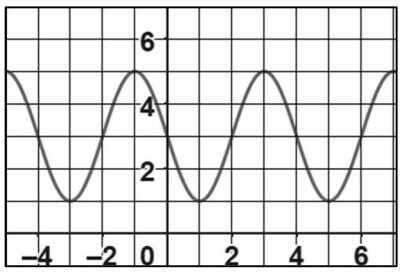
37. A ball is dropped from a height of 82 feet. The ball will retain approximately 75% of its height after each bounce. Which bounce is the first bounce in which the height of the bounce is less than five feet?

(A) 7 (B) 8 (C) 9 (D) 10 (E) 11

38. Assume the average life expectancy of women in Tampa is 83.7 years and the average life expectancy for men living in Tampa is 81.2 years. If the average of all men and women in Tampa is 82.8 years, what is the number of women in Tampa divided by the number of men? (nearest hundredth)

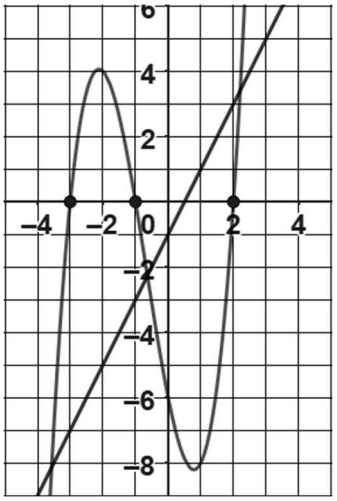
(A) 1.74 (B) 1.75 (C) 1.76 (D) 1.77 (E) 1.78

39. Consider the graph of $f(x)$ as shown to the right. What is the value of $f(0.5)$? (nearest hundredth)



(A) 1.55 (B) 1.56 (C) 1.57
(D) 1.58 (E) 1.59

40 – 42. Consider the graph of the cubic function, $f(x)$, and the linear function $g(x) = ax + b$ such that a and b are integers.



40. What is the largest distance between any two coordinates where $g(x) = f(x)$? (nearest hundredth)

(A) 12.91 (B) 12.93 (C) 12.95 (D) 12.97 (E) 12.99

41. What is the y-intercept of the line tangent to $f(x)$ at $x = 1$? (nearest tenth)

(A) $(0, -10.1)$ (B) $(0, -10.0)$ (C) $(0, -9.9)$
(D) $(0, -9.8)$ (E) $(0, -9.7)$

42. What is the area of the region bounded by the graphs of $f(x)$ and $g(x)$ in the fourth quadrant? (nearest hundredth)

(A) 12.42 (B) 12.48 (C) 12.55 (D) 12.61 (E) 12.67

43. Consider the ellipse $2x^2 + 3y^2 = 12$. The ellipse is the base of a solid that has semicircular cross-sections perpendicular to the x-axis. What is the volume of the solid? (nearest tenth)

(A) 20.2 (B) 20.3 (C) 20.4 (D) 20.5 (E) 20.6

44 – 45. The number of monkeys bouncing on beds in a mattress store is increasing at a rate of $B(t) = \frac{5\sin t}{t+1} + 5$ monkeys per second from $t = 0$ seconds to $t = 8$ seconds. The manager of the store first noticed the monkeys when there were 22 monkeys at $t = 0$.

44. How many monkeys are bouncing on beds at $t = 8$? (nearest integer)

(A) 63 (B) 64 (C) 65 (D) 66 (E) 67

45. At $t = 8$, the monkeys stop jumping on the beds because the manager is able to start getting rid of them at a rate of $E(t) = t\sin^2(0.2t)$ monkeys per second for $t \geq 8$. When $t =$ ___, there will only be one monkey left. Assume there is a whole number of monkeys at $t = 8$. (nearest hundredth)

(A) 20.62 seconds (B) 20.66 seconds (C) 20.70 seconds (D) 20.74 seconds (E) 20.78 seconds

46. What is the average value of the function $f(x) = \frac{e^{3x}}{x^x}$ on the interval $[0.5, 1.2]$? (nearest tenth)

(A) 15.7 (B) 15.8 (C) 15.9 (D) 16.0 (E) 16.1

47. Consider the position vector $\langle 6t-2, t^2-12t \rangle$ for $t \geq 0$. What is the magnitude of the velocity vector when $t = 4$? (nearest hundredth)

(A) 7.19 (B) 7.21 (C) 7.23 (D) 7.25 (E) 7.27

48. Find the length of the arc from $\theta = \frac{\pi}{3}$ to $\theta = \frac{2\pi}{3}$ for the polar curve $r = 3 - 2\cos(5\theta)$. (nearest hundredth)

(A) 7.13 (B) 7.15 (C) 7.17 (D) 7.19 (E) 7.21

49. Find the sum of the series given by $2 - \frac{8}{3!} + \frac{32}{5!} - \frac{128}{7!} + \ldots$. (nearest thousandth)

(A) 0.907 (B) 0.908 (C) 0.909 (D) 0.910 (E) 0.911

50. Newton's Law of Cooling states the rate of change in the temperature of an object is proportional to the difference between the temperature of the object and the temperature of the surrounding substance. A boiled egg was taken out of the water when it had reached $170°F$ and placed in a water bath that remains a constant $42°F$. After 12 minutes the egg was down to $105°F$. The temperature of the egg should reach $60°F$ after another _____ minutes. (nearest tenth)

(A) 20.8 minutes (B) 21.0 minutes (C) 21.2 minutes (D) 21.4 minutes (E) 21.6 minutes

51. What is the interval of convergence of the series $\sum_{n=1}^{\infty} \frac{2(x+1)^n}{n \cdot 3^n}$?

(A) $(-2,4]$ (B) $(-2,4)$ (C) $[-2,4)$ (D) $(-4,2]$ (E) $[-4,2)$

Week	1	2	3	4	5	6
Miles Run	28.6	32.4	37.1	41.4	45.2	49.6

52 – 54. Sarah is training for a marathon and keeps track of how many miles per week she runs. She notices a strong, positive, linear relationship in the data.

52. What is the sum of the median and mean of the number of miles Sarah ran each week? (nearest tenth)

(A) 77.9 (B) 78.0 (C) 78.1 (D) 78.2 (E) 78.3

53. Sarah analyzed the data by calculating a LSRL. How many miles does Sarah expect to run during the seventh week? (nearest tenth)

(A) 53.8 miles (B) 53.9 miles (C) 54.0 miles (D) 54.1 miles (E) 54.2 miles

54. Calculate the standard deviation of the residuals to find the approximate size of a typical prediction error. (nearest hundredth)

(A) 0.237 (B) 0.239 (C) 0.241 (D) 0.243 (E) 0.245

55. Suppose 63% of students at Dulles like to drink milk. Of those, 28% like grapes. Of the 37% who do not like to drink milk, 42% like grapes. Given that a randomly selected student likes grapes, what is the probability that this student likes to drink milk? (nearest hundredth)

(A) 0.47 (B) 0.49 (C) 0.51 (D) 0.53 (E) 0.55

56 – 57. Principal Berry thinks the new communication system she set up for teachers increased their ability to focus on the students, which increased the student average grade. Last year, the average grade was 81.6. She took a random sample of 25 of this year's students and found their mean score to be 82.8 with a standard deviation of 2.6. Assume the grades of the students are approximately normally distributed. Principal Berry had the statistics class perform an appropriate statistical test on the data.

56. The proper null hypothesis is $H_0 : \mu = 81.6$ and the proper alternative hypothesis is _____.

(A) $H_a : \mu > 82.8$ (B) $H_a : \mu < 82.8$ (C) $H_a : \mu > 81.6$ (D) $H_a : \mu < 81.6$ (E) $H_a : \mu \neq 81.6$

57. At the $\alpha = 0.05$ level, the statistics class rejected H_0 because the P-value was _____. (nearest ten-thousandth)

(A) 0.0142 (B) 0.0144 (C) 0.0146 (D) 0.0148 (E) 0.0150

58. Anthony loves to play disc golf and throw the disc as far as he can. The distribution of the distance of drives by Anthony in his career is approximately normal with a mean of 370 feet and a standard deviation of 20 feet. What percentage of his drives are over 400 feet? (nearest hundredth)

(A) 6.68% (B) 6.70% (C) 6.72% (D) 6.74% (E) 6.76%

59. A wildlife sanctuary recently brought in 900 monkeys of various weights. Previously, the sanctuary had no monkeys in it. The distribution of weights of the monkeys is approximately normal with a mean weight of 14.5 pounds and a standard deviation of 1.2 pounds. A researcher collected a random sample of 20 monkeys just after they were released into the sanctuary. What is the probability that the mean weight of the monkeys the researcher collected is greater than 15 pounds? (nearest thousandth)

(A) 0.029 (B) 0.031 (C) 0.033 (D) 0.035 (E) 0.037

60. The local newspaper conducted a survey to find the proportion of high school seniors in their town plan on going to college. They intend to use a 92% confidence interval with a margin of error of no more than 2.0%. Of the following, what is the smallest sample size that the newspaper should use?

(A) 1,830 (B) 1,860 (C) 1,890 (D) 1,920 (E) 1,950

2025 – 2026
Virtual Challenge Meet #4
Mathematics – Student Answer Sheet

Contestant Name_____ Grade_____

Score 1: _____	Score 2: _____
Score 3: _____	**Final Score: _____**

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2025 – 2026 Virtual Challenge Meet #4 – Mathematics Key

1. D	**21. B**	**41. B**
2. D	**22. E**	**42. A**
3. E	**23. A**	**43. D**
4. A	**24. D**	**44. C**
5. B	**25. C**	**45. A**
6. C	**26. E**	**46. B**
7. D	**27. E**	**47. B**
8. E	**28. A**	**48. A**
9. A	**29. A**	**49. C**
10. C	**30. D**	**50. C**
11. D	**31. B**	**51. E**
12. E	**32. D**	**52. E**
13. B	**33. A**	**53. A**
14. B	**34. B**	**54. B**
15. A	**35. C**	**55. D**
16. B	**36. C**	**56. C**
17. C	**37. D**	**57. E**
18. D	**38. E**	**58. A**
19. E	**39. E**	**59. B**
20. C	**40. A**	**60. D**